

EC 9540: HUMAN COMPUTER INTERACTION
MINI GROUP PROJECT
LAB 02



YamuLK

**A Smart Travel Planning Platform for Sri Lankan
Domestic Travelers**

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Introduction

Following Lab 01, a questionnaire was conducted among potential users to better understand the needs and expectations of Sri Lankan domestic travelers. The collected responses were analyzed to identify the common problems users face while planning local trips and to determine the essential features required in the proposed system.

The findings were used to identify the core user task, define the user's primary goal, break it down into smaller sub-tasks, and formulate a hierarchical task plan.

Target Users

The target users of **YamuLK** are Sri Lankan residents who travel within Sri Lanka for leisure, family vacations, educational trips, religious visits, or weekend getaways.

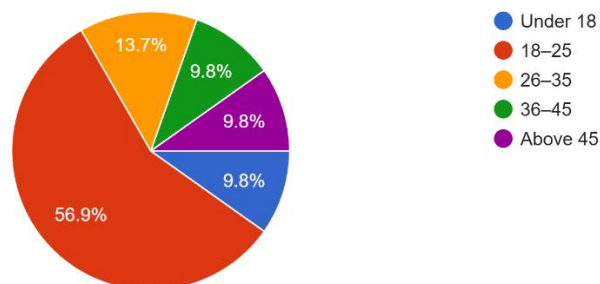
User Categories:

- University students
- Young professionals
- Families
- Groups of friends
- Solo travelers
- Local tourists exploring new destinations

Summary of Questionnaire Findings

The survey revealed several important insights into the travel habits and expectations of Sri Lankan domestic travelers.

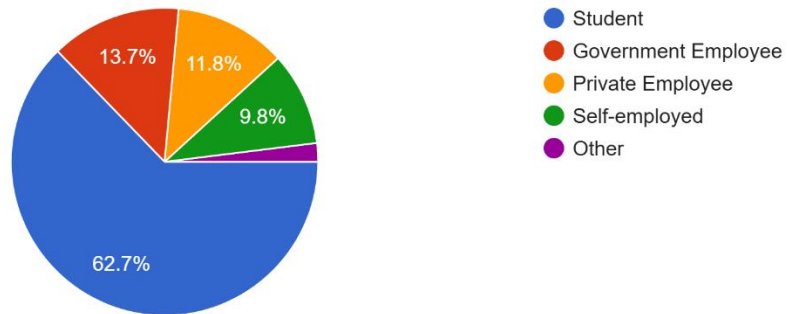
1. What is your age group?
51 responses



- Mostly between 18–25 years old

2. What is your occupation?

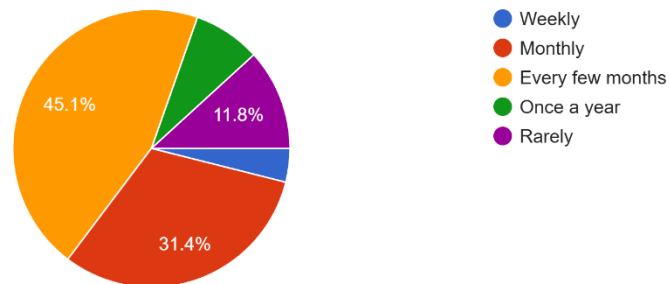
51 responses



- Majority are students

4. How often do you travel within Sri Lanka?

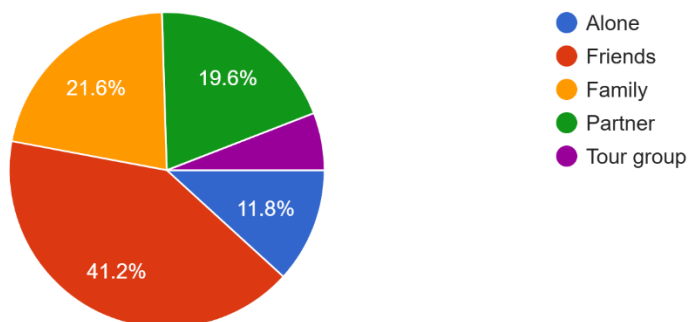
51 responses



- Most respondents travel every few months, showing that domestic travel is a regular activity for many users.

5. Who do you usually travel with?

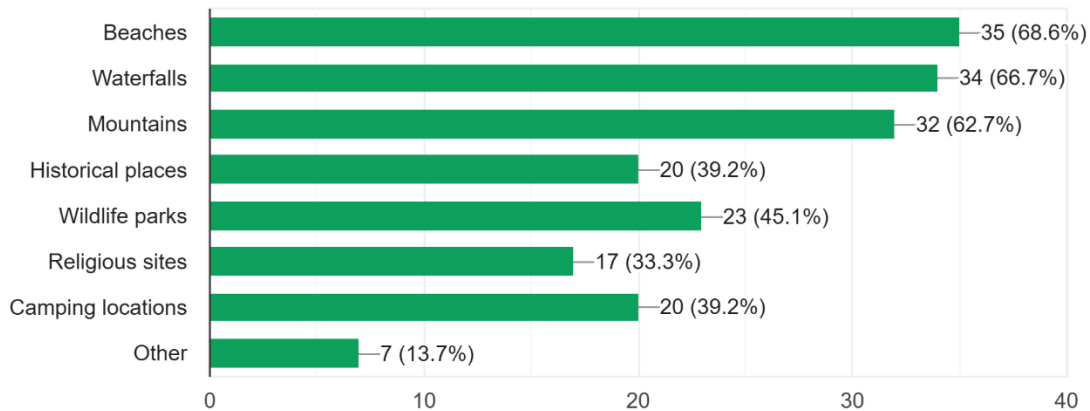
51 responses



- Most participants prefer travelling with friends, while others commonly travel with family or partners.

6. What type of destinations do you usually prefer? (Select all that apply.)

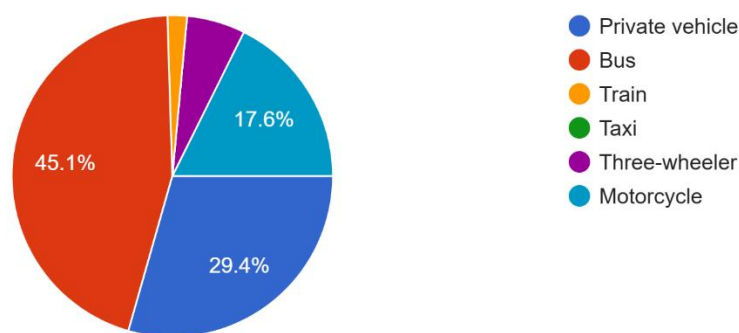
51 responses



- Beaches, waterfalls, and mountains were the most popular destination types among respondents.

7. Which transportation method do you mostly use?

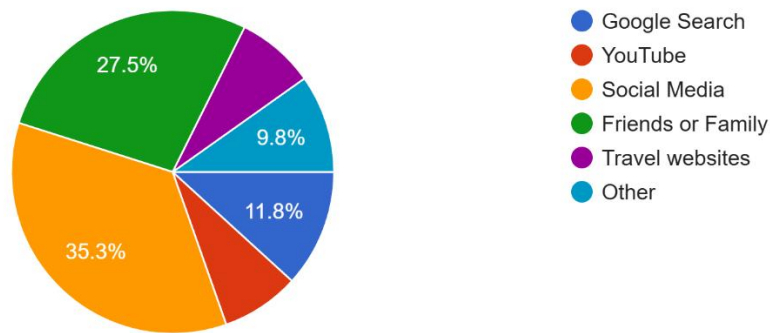
51 responses



- Buses were the most commonly used mode of transportation, followed by private vehicles and motorcycles.

8. How do you usually plan your trips?

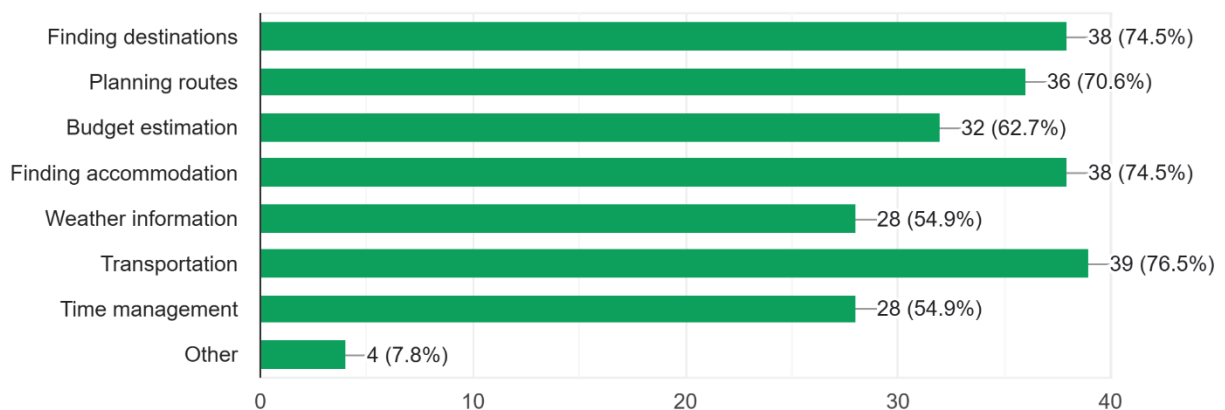
51 responses



- Most respondents rely on social media and recommendations from friends or family when planning trips.

9. What difficulties do you usually face while planning a trip? (Select all that apply.)

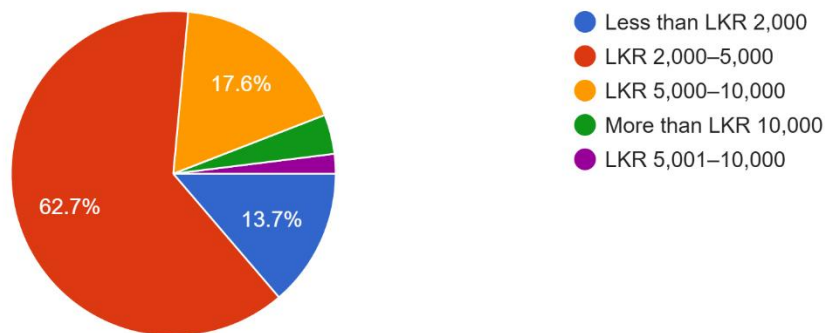
51 responses



- The most common challenges were transportation, finding destinations, accommodation, and planning travel routes.

10. What is your average budget for a one-day trip?

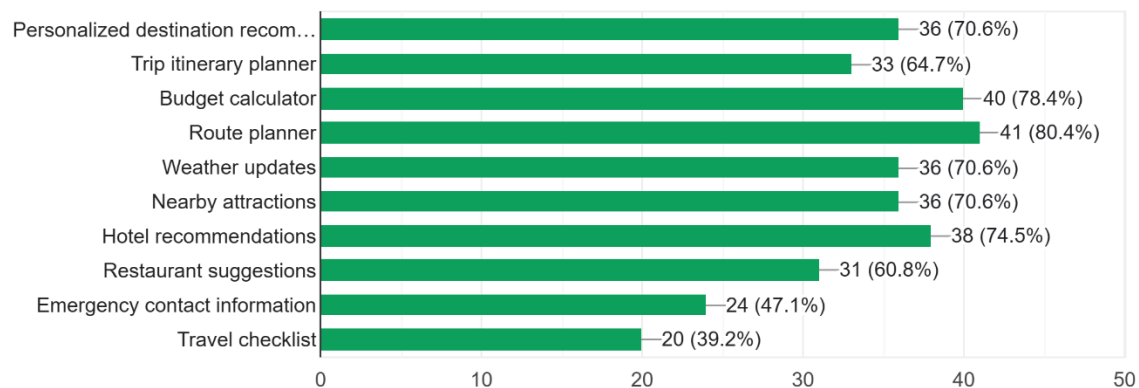
51 responses



- Most respondents preferred a moderate budget range (approximately LKR 2,000–5,000) for a one-day trip.

11. Which features would you like to have in a travel planning application? (Select all that apply.)

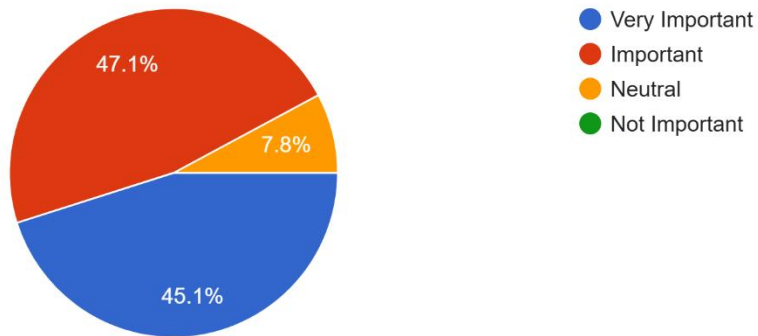
51 responses



- Budget calculation, route planning, hotel recommendations, weather updates, and nearby attraction suggestions were the most requested features.

12. How important is a budget estimation feature in a travel planning application?

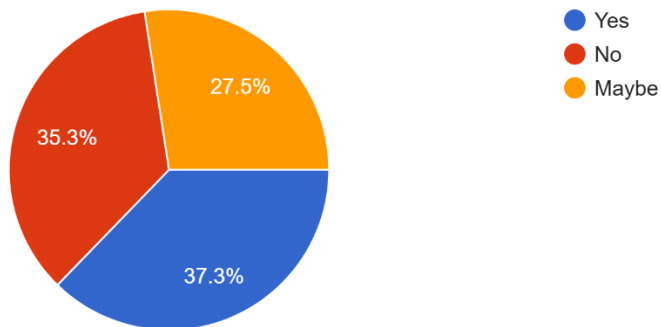
51 responses



- Almost all respondents considered a budget estimation feature to be important or very important.

13. Would you use a mobile application that automatically recommends travel plans based on your interests and budget?

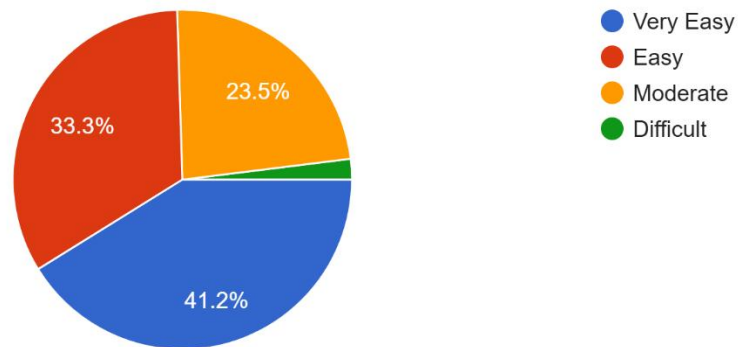
51 responses



- Most respondents were **positive or open to using** an application that provides personalized travel recommendations.

14. How easy should such an application be to use?

51 responses



- Most respondents expected the application to be easy or very easy to use.

15. Do you have any additional suggestions or features you would like to see in YamuLK?

9 responses

✦ Summarise responses

user friendly interface

add dark mode also

Best time management

User reviews for hidden places

If the map also there it would be nice

-

Include offline maps.

Add a feature to split bills with friends.

Large font size for easy reading

User Requirements and Expectations

Based on the questionnaire findings, the following requirements were identified.

Functional Requirements

The system should allow users to:

- Search tourist destinations
- Receive personalized destination recommendations
- Calculate estimated travel costs
- Generate travel itineraries
- View optimized travel routes
- Find nearby hotels and accommodations
- Discover nearby restaurants
- View nearby attractions
- Check weather forecasts
- Save favorite destinations
- Access emergency contact information
- Maintain a travel checklist

Non-Functional Requirements

The application should:

- Be simple and easy to use
- Have a clean and attractive interface
- Provide accurate travel information
- Respond quickly
- Be mobile-friendly
- Ensure reliability and security
- Support future scalability

Core Task Performed by the Target User

The core task identified from the survey is:

- *Planning a complete domestic trip within Sri Lanka using a single platform.*

This includes,

- Discovering destinations
- Planning routes
- Estimating expenses
- Finding accommodation
- Organizing the trip efficiently

Main Goal of the User

To plan an enjoyable, affordable, and well-organized trip within Sri Lanka quickly and conveniently.

Breakdown of the Main Goal into Sub-Tasks

1. Access the application.
2. Search for travel destinations.
3. Explore destination information.
4. Compare available options.
5. Select a destination.
6. Estimate travel expenses.
7. Generate an itinerary.
8. Find accommodation and restaurants.
9. Check weather conditions.
10. Plan the travel route.
11. Save or share the travel plan.

Actions Required to Complete Each Sub-Task

Sub-Task	User Actions
Access Application	Open YamuLK and log in or continue as guest
Search Destination	Enter destination name or apply filters
Explore Information	Read descriptions, images, ratings, and reviews
Compare Destinations	Compare costs, distance, attractions, and facilities
Select Destination	Choose preferred travel location
Estimate Budget	Enter trip details and calculate expected expenses
Generate Itinerary	Select travel dates and preferences
Find Accommodation	View nearby hotels and restaurants
Check Weather	View weather forecast before travelling
Plan Route	Display optimized navigation route
Save/Share Plan	Save itinerary or share with travel companions

Hierarchical Task Plan (HTA)

Plan a Domestic Trip

1. Access YamuLK

- 1.1 Open application
- 1.2 Login/Register
- 1.3 Continue as Guest

2. Search Destination

- 2.1 Enter destination
- 2.2 Apply filters
- 2.3 View recommendations

3. Evaluate Destination

- 3.1 View destination details
- 3.2 Compare destinations
- 3.3 Select destination

4. Plan Trip

- 4.1 Estimate budget
- 4.2 Generate itinerary
- 4.3 Find accommodation
- 4.4 Find restaurants
- 4.5 View nearby attractions
- 4.6 Check weather

5. Travel

- 5.1 Plan route
- 5.2 Start navigation

6. Finalize

- 6.1 Save travel plan
- 6.2 Share travel plan

Conclusion

The analysis of the first 50 questionnaire responses demonstrates that Sri Lankan domestic travelers need a centralized travel planning platform that simplifies the entire planning process. The survey identified transportation, accommodation, destination discovery, route planning, and budget estimation as the major challenges faced by travelers. Users strongly prefer features such as budget calculation, route planning, weather updates, hotel recommendations, and personalized destination suggestions. These findings provide a strong foundation for designing the YamuLK system and will guide the interface design and implementation in the next stages of the project.